

Logarithmic Functions

I. Express in terms of $\ln 2$ and $\ln 3$ the numbers:

- a) $\ln 24$ b) $\ln 144$ c) $\ln\left(\frac{4}{9}\right)$
d) $\ln(3\sqrt{2})$ e) $\ln\left(8\sqrt[3]{\frac{2}{3}}\right)$ f) $\ln 27 - \ln 3 - \ln 2 + \ln 8$
g) $\ln 42 - \ln 14 + \ln \frac{2}{3}$ h) $\ln\left(\frac{8}{27}\right) + 2\ln 4 + 3\ln 3$

II. Simplify:

- a) $\ln(\sqrt[3]{e^2}) - 3 + 2\ln(\sqrt{e^5})$ b) $\ln(2 + \sqrt{3})^{30} + \ln(2 - \sqrt{3})^{30}$
c) $\ln(3x^2 - 9x) + \ln\left(\frac{1}{3x}\right)$ d) $3\ln(\sqrt[3]{x^2 - 1}) - \ln(x + 1)$

III. Determine the derivative of the following functions:

- a) $f(x) = \ln 3x$ b) $f(x) = \ln(2x - 1)$ c) $f(x) = \ln\left(\frac{3}{x}\right)$ d) $f(x) = \ln(\ln x)$
e) $f(x) = \ln\left(\frac{x^2 + 1}{x + 2}\right)$ f) $f(x) = \frac{\ln x - 1}{x + 1}$ g) $f(x) = \ln \sqrt{x^2 - 3x - 4}$
h) $f(x) = \ln \left| \frac{x - 2}{2x + 1} \right|$ i) $f(x) = \frac{\ln x}{1 + \ln x}$ j) $f(x) = \ln \frac{(x^2 + 1)^5}{\sqrt{1 - x^2}}$
k) $f(x) = \ln(1 + \sqrt{x})$ l) $f(x) = \sqrt{1 + \ln x}$ m) $f(x) = \frac{\ln x + 1}{\ln x - 1}$

IV. Solve the following equations inequations and systems:

- 1) $\ln x = 1$ 2) $\ln 2x = 0$ 3) $\ln x = -1$
4) $\ln x = 3$ 5) $\ln(2x + 1) = -2$ 6) $(x + 2)\ln(x + 2) = 0$
7) $\ln(x + 1) - \ln(x + 3) = \ln 2$ 8) $\ln(x^2 - 2x - 3) = \ln(x + 7)$
9) $\ln(x^2 + 5x + 6) = \ln(x + 1)$ 10) $\ln(x^2 - x + 4) - \ln(3x - 1) = \ln 2$
11) $\ln(x - 3) = \ln(x + 7) - \ln(x + 1)$ 12) $\ln(2x + 1) + \ln 3 = \ln(x^2 + 7x + 1)$
13) $\ln(x - 2) - \ln(x - 3) \leq \ln 2$ 14) $\ln^2 x - 3\ln x + 2 < 0$
15) $\ln^2 x - \ln x - 2 \leq 0$ 16) $(\ln x)^2 - 5\ln x + 6 > 0$
17) $\begin{cases} 2\ln x + \ln y = 7 \\ 3\ln x - 5\ln y = 4 \end{cases}$ 18) $\begin{cases} x^2 + y^2 = 130 \\ \ln x + \ln y = \ln 63 \end{cases}$ 19) $\begin{cases} x + y = 30 \\ \ln x + \ln y = 3\ln 6 \end{cases}$

V. Calculate the following limits:

- 1) $\lim_{x \rightarrow 0} x \ln 3x$ 2) $\lim_{x \rightarrow 0} x^2 \ln x$ 3) $\lim_{x \rightarrow 0} \frac{\ln 3x}{x^2}$

- 4) $\lim_{x \rightarrow +\infty} \frac{\ln 3x}{x^2}$ 5) $\lim_{x \rightarrow +\infty} (\ln x - x)$ 6) $\lim_{x \rightarrow 0} \frac{\ln x - 2}{\ln x - 1}$
- 7) $\lim_{x \rightarrow +\infty} \frac{\ln x - 2}{\ln x - 1}$ 8) $\lim_{x \rightarrow 0} \frac{\ln(1+x^2)}{x}$ 9) $\lim_{x \rightarrow +\infty} \frac{\ln(1+x^2)}{x}$
- 10) $\lim_{x \rightarrow 3} \frac{\ln x - \ln 3}{x - 3}$ 11) $\lim_{x \rightarrow +\infty} \frac{\ln x}{x^2 + 1}$ 12) $\lim_{x \rightarrow 0} \frac{x - \ln x}{x}$
- 13) $\lim_{x \rightarrow +\infty} \frac{x - \ln x}{x}$ 14) $\lim_{x \rightarrow +\infty} \left(\frac{\ln x}{x^2} - x^2 \right)$ 15) $\lim_{x \rightarrow 0} \frac{\ln(1+x)}{x^2 + x}$
- 16) $\lim_{x \rightarrow +\infty} (x^3 - 2x - \ln x)$ 17) $\lim_{x \rightarrow +\infty} (\ln(2x+5) - \ln(3x-1))$
- 18) $\lim_{x \rightarrow 0} x \ln \left(1 + \frac{1}{x} \right)$ 19) $2 \lim_{x \rightarrow +\infty} x \ln \left(1 + \frac{1}{x} \right)$ 20) $\lim_{x \rightarrow 0} x^2 (\ln x + 1)$
- 21) $\lim_{x \rightarrow -\infty} \frac{x \ln |x|}{\ln(1-x)}$ 22) $\lim_{x \rightarrow e} \frac{\ln x - 1}{x - e}$

VI. Consider the functions $f(x) = (x-1) + \frac{\ln(x-1)}{x-1}$ and $g(x) = x + \ln x - \ln(x-1)$

a) Find the $\lim_{x \rightarrow 1^+} f(x)$, $\lim_{x \rightarrow 1^+} g(x)$, $\lim_{x \rightarrow +\infty} (f(x) - x + 1)$ and $\lim_{x \rightarrow +\infty} (g(x) - x)$.

b) Deduce the asymptotes of the graphs of f and g .

VII. Calculate the integral of the following:

- 1) $\int \frac{1}{1+x} dx$ 2) $\int \left(1 + \frac{1}{3x-1} \right) dx$ 3) $\int \frac{x}{x^2+2} dx$ 4) $\int \frac{x}{1-x^2} dx$
- 5) $\int \frac{\ln x}{x} dx$ 6) $\int \frac{\ln^2 x}{x} dx$ 7) $\int \frac{1}{x \ln x} dx$ 8) $\int \frac{dx}{x \ln^2 x}$
- 9) $\int \frac{dx}{x(1+\ln x)}$ 10) $\int \frac{dx}{x(1+\ln x)^2}$ 11) $\int \frac{x+1}{x^2+2x+3} dx$
- 12) $\int \frac{\cos x}{2 \sin x + 3} dx$ 13) $\int \frac{1 + \tan^2 x}{\tan x} dx$ 14) $\int \frac{1}{\cos^2 x (1 + 3 \tan x)} dx$
- 15) $\int \frac{\sqrt{1+\ln x}}{x} dx$ 16) $\int \frac{x dx}{(x^2+1) \ln(x^2+1)}$ 17) $\int \frac{x^2}{2x-1} dx$
- 18) $\int \frac{3x^2 - 4x + 2}{x-1} dx$ 19) $\int \frac{dx}{x(x^2+1)}$ 20) $\int_1^2 \frac{1}{1+x} dx$ 21) $\int_{-3}^{-2} \frac{dx}{x}$
- 22) $\int_2^4 \frac{dx}{x \ln^2 x}$ 23) $\int_0^{\frac{\pi}{2}} \tan x dx$ 24) $\int_0^{\frac{\pi}{3}} \frac{\sin x dx}{1-4 \cos x}$ 25) $\int_6^7 \frac{x dx}{x^2-25}$

$$\begin{array}{llll}
26) \int_1^2 \frac{xdx}{4x^2 - 1} & 27) \int_1^2 \frac{dx}{x^2 + 2x} & 28) \int_1^2 \frac{dx}{x^2 + 3x} & 29) \int_{-2}^0 \frac{dx}{x^2 + x - 6} \\
30) \int_0^1 \frac{xdx}{x^2 + 3x + 2} & 31) \int_{-1}^1 \ln\left(\frac{2-x}{2+x}\right) dx & &
\end{array}$$

VIII.

- 1) a) Study the sense of variations of $g(x) = x - 1 + \ln x$ over $]0, +\infty[$
b) Verify that $g(1) = 0$. Deduce the sign of g .
- 2) Consider the function $f(x) = \frac{x-1}{x} \ln x$. Let (C) be its representative curve.
 - a) Show that for all $x \in]0, +\infty[$ we have $f'(x) = \frac{g(x)}{x^2}$
 - b) Determine the limits of $f(x)$ as x tends to 0 and $+\infty$.
 - c) Set up the table of variations of f .
- 3) Study, according to the values of x , the sign of $(f(x) - \ln x)$.
Deduce the relative position of (C) with respect to curve (H) of the function $h(x) = \ln x$.
- 4) Draw the curves (H) and (C) .
- 5) Find the area bounded by (H) , (C) and the two vertical lines of equations $x = 1$ and $x = e$.

IX.

- 1) Study the sense of variations of $h(x) = x - \ln x$ over $]0, +\infty[$ and deduce the its sign.
- 2) Let f be the function defined by $f(x) = \frac{x}{x - \ln x}$. Let (C) be its representative curve.
 - a) Determine the domain of definition of f .
 - b) Calculate $\lim_{x \rightarrow 0^+} f(x)$ and $\lim_{x \rightarrow +\infty} f(x)$. Deduce an asymptote to (C) .
 - c) Calculate $f'(x)$. Set up the table of variations of f .
 - d) Find the equation of the tangent (d) to (C) at a point A of abscissa 1. Study the relative position of (C) and (d) .
 - e) Justify that $\lim_{x \rightarrow 0^+} f'(x) = 0$
 - f) Draw (d) and (C) .

X.

- Consider the function f defined over $]0, +\infty[$ by $f(x) = x^2 + 2 - 2\ln x$. Let (C) be its representative curve.
- 1) Study the variations of f and set up the table of variations of f .
Deduce the sign of $f(x)$ for $x > 0$.
 - 2) Study the variations of the function g defined over $]0, +\infty[$ by $g(x) = x + \frac{2 \ln x}{x}$
 - 3) Verify that (G) admits an asymptote (d) of equation $y = x$ at $+\infty$.
 - 4) Draw (d) and (G) .
 - 5) Determine the point of (G) where the tangent is parallel to (d) .
 - 6) Calculate the area of the region bounded by (C) , (d) and the two vertical lines of equations $x = 1$ and $x = e$.
 - 7) Solve $x + \frac{2 \ln x}{x} > e + \frac{2}{e}$

- XI.** A- 1) Determine the limits of $g(x) = -x^2 + 1 - \ln x$ as x tends to 0 and $+\infty$.
 2) Study the variations of g over $]0, +\infty[$.
 3) Calculate $g(1)$. Deduce the sign of g over $]0, +\infty[$.

B- Consider the function $f(x) = -\frac{1}{2}x + 1 + \frac{\ln x}{2x}$

- 1) Determine $f'(x)$ in terms of $g(x)$.
 2) Calculate $\lim_{x \rightarrow 0^+} f(x)$, $\lim_{x \rightarrow +\infty} f(x)$ and $\lim_{x \rightarrow +\infty} \left(f(x) + \frac{1}{2}x - 1 \right)$. Deduce the asymptotes of (C) of the function f .
 3) Set up the table of variations of f .

C- Discuss the sign of $\left(f(x) + \frac{1}{2}x - 1 \right)$. Deduce the relative position of (C) and (d) :

$$y = -\frac{x}{2} + 1. \text{ Trace } (d) \text{ and } (C).$$

- XII.** A- Study the sense of variation of $g(x) = x^2 - 2\ln x$ over $]0, +\infty[$ and deduce its sign.

B- Consider the function f defined over $]0, +\infty[$ by $f(x) = \frac{x}{2} + \frac{1 + \ln x}{x}$. Let (C) its representative curve.

- 1) Determine the limit of f at 0. Interpret the result graphically.
 2) a) Find $\lim_{x \rightarrow +\infty} f(x)$. Show that the straight line $(D): y = \frac{x}{2}$ is an asymptote to (C) .
 b) Determine the relative position of (D) and (C) . Show in particular that (D) cuts (C) at a point A to be determined.
 3) Study the variation of f .
 4) Show that there exists one and only one point B of (C) where the tangent (T) to (C) is parallel to (D) . Find the coordinates of B .
 5) Show that the equation $f(x) = 0$ has a solution α such that $0.34 < \alpha < 0.35$.
 6) Draw (D) , (T) and (C) .

- XIII.** A- Consider the function g defined over $]0, +\infty[$ by $g(x) = -x^2 - 4 + 4\ln x$. Study the sense of variations of g and deduce its sign over $]0, +\infty[$.

B- Consider over $]0, +\infty[$ the function f defined by $f(x) = 2 - x - 4 \frac{\ln x}{x}$. Let (C) be its representative curve.

- 1) Determine the limits of f at 0 and $+\infty$. Interpret geometrically the result.
 2) a) Show that the straight line $(D): y = 2 - x$ is an asymptote to (C) .
 b) Determine the position of (C) with respect to (D) .
 3) Study the variations of f and set up its table of variations.
 4) Show that there exists one point B of (C) where the tangent (T) to (C) is parallel to (D) . Determine the coordinates of B .
 5) Show that $f(x) = 0$ admits a solution α such that $1 < \alpha < 1.5$.

6) Draw (D) , (T) and (C) .

7) Consider over $]0, +\infty[$ the function h defined by $h(x) = (\ln x)^2$. Deduce the primitive of f .

XIV. A- Consider the function f defined over $]0, +\infty[$ by $f(x) = 1 + \frac{\ln x}{x}$. Let (C) be its representative curve.

1) Determine the limits of f at 0 and $+\infty$. Interpret geometrically the results.

2) a) Study the sense of variations of f .

b) Show that the equation $f(x) = 0$ has over $\left[1, \frac{1}{e}\right]$ has one solution α . Show that

$0.56 < \alpha < 0.57$ Deduce the sign of f over $]0, +\infty[$

c) Draw (C) .

B- 1) Determine the equation of tangent (D) to (C) at a point of abscissa 1.

2) a) Let g be the function defined over $]0, +\infty[$ by $g(x) = x - x^2 + \ln x$.

b) Find $g'(x)$ and set up the table of variations of g . Give its sign.

c) Show that $f(x) - x = \frac{g(x)}{x}$. Deduce the relative position of (C) and (D) .

d) Consider the domain limited by the x -axis, the curve (C) and (D) . Find its area.

XV. A- Consider the polynomial function $p(x) = 3x^3 - x - 2$

1) Verify that $p(x) = (x - 1)(3x^2 + 3x + 2)$ and deduce its sign.

2) Consider the function g defined over $]0, +\infty[$ by $g(x) = x^3 - x + 1 - 2\ln x$

Show that $g'(x) = \frac{p(x)}{x}$. Deduce the variations of g and its sign.

B- Consider the function f defined over $]0, +\infty[$ by $f(x) = x + 1 + \frac{x + \ln x}{x^2}$. Let (C) be its representative curve. (unit: 2 cm)

1) a) Determine the limit of $f(x)$ as x tends to 0 and $+\infty$. Interpret geometrically.

b) Show that the straight line (d) of equation $y = x + 1$ is an asymptote to (C) .

2) a) Study the variations of the function $h(x) = x + \ln x$

b) Show that the equation $h(x) = 0$ admits a unique solution α such that $0.56 < \alpha < 0.57$. Deduce the sign of $h(x)$.

c) Determine the relative position of (C) with respect to (d) .

3) Study the variations of f and draw (C) .

4) Show that f admits an inverse function f^{-1} . Draw its curve.

5) a) Calculate $\int_1^e \frac{\ln x}{x^2} dx$ (using integration by part)

b) Calculate in cm^2 the area of the domain limited by (C), (d) and the two straight lines $x = 1$ and $x = e$.

XVI. A- Consider the function g defined over $]0, +\infty[$ by $g(x) = -2\ln x - xe + 1$.

- 1) Determine the limit of $g(x)$ as x tends to 0 and $+\infty$.
- 2) Study the variations of the function g .
- 3) Show that the equation $g(x) = 0$ admits one solution α such that $0.6 < \alpha < 0.7$
- 4) Deduce the sign of g over $]0, +\infty[$.

B- Consider the function f defined over $]0, +\infty[$ by $f(x) = \frac{\ln x + xe}{x^2}$

- 1) Determine the limits of f at 0 and $+\infty$.
- 2) Calculate $f'(x)$ in terms of $g(x)$.
- 3) Set up the table of variations of f .
- 4) Show that $f(\alpha) = \frac{1 + \alpha e}{2\alpha^2}$
- 5) Draw (C).

XVII. 1) Show that $2x^2 - x + 1$ is positive for every $x \in \mathbb{R}$, deduce that $2x^2 + 1 > x$.

2) Study the variations of the function $g(x) = \frac{\ln x}{x}$ over $]0, +\infty[$. Deduce that $\ln x < x$.

3) Deduce that $\ln x < 2x^2 + 1$.

4) Consider the function f defined over $]0, +\infty[$ by $f(x) = \frac{2x^2 + \ln x}{x}$. Let (C) be its

representative curve. (unit: 2 cm)

- a) Determine the limits of f at 0 and $+\infty$. Interpret geometrically.
- b) Show that the straight line (D): $y = 2x$ is an asymptote to (C) at $+\infty$.
- c) Study the relative position of (C) with respect to (D).
- d) Determine the point of (C) where the tangent is parallel to (D).
- e) Study the variations of f . Draw (C).
- f) Calculate the area of the region, set of points $M(x, y)$ such that:
 $1 \leq x \leq e$ and $2x \leq y \leq f(x)$.

XVIII. Consider the function f defined over $]0, +\infty[$ by $f(x) = \frac{x}{2} - 1 + \frac{\ln x}{x}$. Designate by (C) its representative curve.

1) Calculate the limits of f at 0 and $+\infty$. Verify that (D): $y = \frac{x}{2} - 1$ is an asymptote to (C).

2) a) Calculate $f'(x)$ and $f''(x)$. Deduce the variations of $f'(x)$.

b) Calculate $f'\left(e^{\frac{3}{2}}\right)$ and deduce the sign of $f'(x)$.

c) Set up the table of variations of f .

3) Draw (D) and (C) .

XIX. A- The adjacent curve (C) represents the graph of the function $f(x) = ax^2 + bx + c + 2\ln(x+1)$ and (T) is the tangent to (C) at the point $A(0, 2)$

1) Calculate in terms of a , b and c , $f'(x)$.

2) Use (C) , determine $f(0)$ and $f'(0)$.

3) If $f'(3) = \frac{1}{2}$, calculate a , b and c .

B- Consider the function f defined over

$$[0; 5] \text{ by } f(x) = \frac{x^2}{2} - 3x + 2 + 2\ln(x+1)$$

1) Determine $f'(x)$ and set up the table of variations of f .

2) Consider the function g defined by $g(x) = (x+1)\ln(x+1) - x$

a) Calculate the derivative of g .

b) Deduce the primitive of the function f .

c) Calculate $\int_0^5 f(x)dx$

